National University of Computer & Emerging Sciences



Computer Vision (AI-4002) | ASSIGNMENT 2

Pedestrian Attribute Recognition with ResNet50

UPAR Challenge 2027, Track 1

Group Members

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Submitted to

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1. Task and approach

Pedestrian Attribute Recognition (PAR) means looking at a cropped image of a person from a surveillance camera and predicting 40 yes/no attributes, such as age group, gender, hair length, upper and lower body clothing type and colour, and accessories like bags, backpacks, glasses and hats. Track 1 of the UPAR Challenge 2027 (Real World Surveillance workshop, WACV 2027) tests this across three public datasets, so the model must work under domain shift between cameras and datasets.

Our approach: we fine-tuned an ImageNet-pretrained ResNet50 for multi-label classification with 40 sigmoid outputs, handled class imbalance in the loss, and then tuned a separate decision threshold for each attribute on the validation set. The final model is packaged as a code submission for Codabench, where it scored 0.7406 on the hidden test set.

2. Dataset and metric

We used the official UPAR 2027 development data. The training split has 97,669 images (PA100k 79,001, PETA 8,668, Market1501 10,000) and the validation split has 33,407 images (Market1501 16,458, PA100k 9,986, PETA 6,963). The data is heavily imbalanced: 20 of the 40 attributes are positive in less than 5% of validation images.

The challenge score is the harmonic mean of label-based mean accuracy (mA, the average of true positive rate and true negative rate per attribute) and instance-based F1. It is computed for each dataset separately and then averaged. As a reference, the official attribute prior baseline, which predicts the same prior probability for every image, scores 54.28.

3. Method

- **Model:** torchvision ResNet50 (ImageNet weights). The final layer is replaced with Dropout(0.2) and Linear(2048, 40).
- **Input:** 256 x 128 crops (pedestrians are tall), ImageNet normalisation.
- **Loss:** binary cross entropy with a higher weight for positive labels of rare attributes (sqrt of negative/positive ratio, clipped to 1 to 8), masking of missing labels, and label smoothing 0.02.
- **Augmentation:** horizontal flip, colour jitter and random erasing (to simulate occlusion). We used this lighter set because Colab has only 2 CPU cores.
- **Training:** AdamW (lr 3e-4, weight decay 0.05), OneCycle schedule, batch size 64, mixed precision, EMA of weights (0.999), gradient clipping, 3 epochs on a Google Colab T4 GPU (about 7 minutes per epoch, 21 minutes in total).
- **Inference:** flip test time augmentation (average of normal and mirrored image), then a per-attribute threshold found by coordinate search over 0.1 to 0.9 on the validation set.

4. Evaluation

Table 1 shows the validation results. Our model is about 32 points above the prior baseline. Threshold tuning adds about 1 point, mostly by raising mA on rare attributes, while F1 drops slightly.

Method	Market1501	PA100k	PETA	Mean mA	Mean F1	Score
Prior baseline	58.17	53.28	51.40	52.50	56.25	54.28
ResNet50 @ 0.5	86.46	84.24	86.08	84.73	86.47	85.59
ResNet50 + tuned thresholds	**87.10**	**85.72**	**86.93**	**87.21**	**85.96**	**86.58**

Table 1: Validation scores (%) per dataset and averaged.

Training progress. The validation score (threshold 0.5, EMA weights) rose from 81.31 after epoch 1 to 84.86 after epoch 2 and 85.40 after epoch 3, while the training loss fell from 0.302 to 0.160. The gain is getting smaller, so a few more epochs would help a little but not a lot.

Is threshold tuning just overfitting? To check, we tuned thresholds on one half of the validation set and tested on the other half. The gain stayed at +0.83 and +0.90 points, so the improvement is real and not only memorising the validation set.

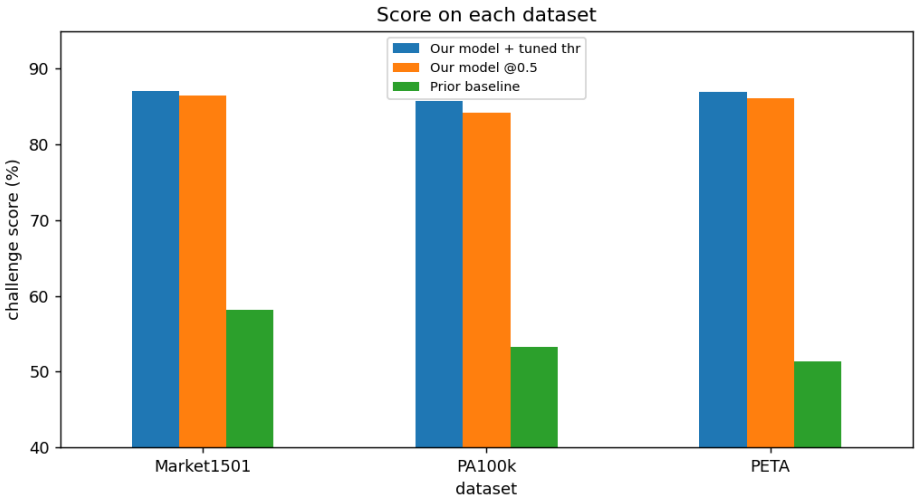


Figure 1: Score on each dataset for the prior baseline, our model at 0.5, and with tuned thresholds.

5. Challenges

- **Rare attributes:** sunglasses (2.2% positives) has the lowest mA (63.9) with a true positive rate of only 29%. Young age (1.1%), bald (1.3%) and LowerBody-Color-Other (3.7%) are also below 78 mA. The model learns to say "no" for these.
- **Small and subtle details:** accessories are the weakest group (mA 79.2, F1 61.9). Bags and glasses cover only a few pixels in a 256 x 128 crop. Lower body colours have the lowest F1 (51.1) because colours are confused under different lighting.
- **Domain shift:** PA100k is the hardest dataset (85.72) even though it is most of the training data, while Market1501 and PETA score above 86.9. Each dataset has its own cameras, resolution and annotation style. On the hidden test set, which comes from a new domain, the score falls to 74.06, about 12.5 points below validation.
- **Practical issues:** the official download script crashed when run twice on Colab, the Colab runtime can disconnect, and with only 2 CPU cores the heavy augmentations made the GPU wait. We added checkpoint saving every epoch with a resume option and used lighter augmentation.

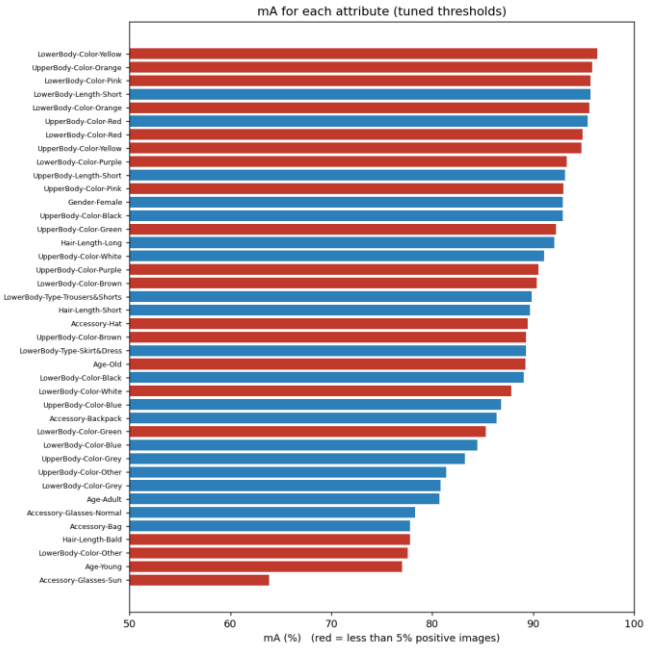


Figure 2: mA per attribute (red = less than 5% positive images). Rare attributes are clearly the weakest.

6. Possible improvements

- Train longer (8 to 12 epochs) and use the full augmentation set (random low resolution, blur, affine).
- Use a stronger backbone such as ConvNeXt or a Vision Transformer, or an ensemble of several models.
- Use a loss designed for imbalance, such as focal loss or asymmetric loss, to improve rare attributes.
- Higher input resolution or attention on body parts to catch small accessories like glasses and bags.
- Domain generalisation tricks (style augmentation, dataset-balanced sampling) to help on the hidden test data.

7. Challenge submission

We packaged the model as a Codabench code submission (run.py with load_model, predict_image and predict_batch, plus fp16 weights, 43.8 MB). Before uploading, we tested the package offline with 15 checks: zip layout, no internet use, output format, attribute order, different input types, and identical decisions to the unpacked model. All 15 passed.

Codabench result: our ResNet50 scored 0.7406 on the hidden test set (submission ID 1002478, submitted on 19 October 2026). The first upload failed because the platform passes image paths relative to its data folder, so we updated run.py to find the image folder automatically; the second upload finished successfully (Figure 3).

Validation vs test: the score drops from 86.58 on validation to 74.06 on the hidden test set. The test images come from a domain the model never saw during training (for example MEVID), so this drop shows how strong the domain shift is and why domain generalisation is the most important next step.

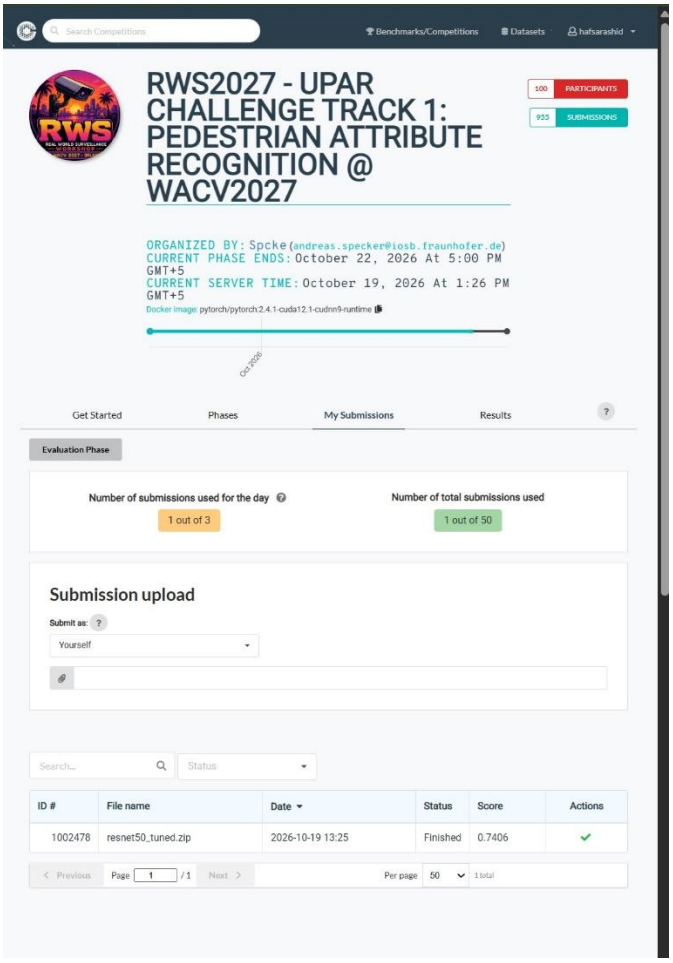


Figure 3: Screenshot of official submission

References

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